Saipranav Venkatakrishnan

847-348-0250 | saipranavvk@gmail.com | linkedin.com/in/saipranav-venkatakrishnan | github.com/Sai-Pra

Education

University of Illinois Urbana-Champaign

B.S. in Computer Engineering — August 2022 – May 2026 (GPA: 3.85)

M.S. in Computer Engineering — August 2026 – May 2028 (GPA: 4.00)

Skills & Tools: SystemVerilog, Verilog, C/C++, CUDA C++, Python; Synopsys Design Compiler, VCS, Verdi, Verilator, Yosys, Spyglass Linter, CDC/RDC Analysis, Nvidia Nsight Compute/Systems, Git, PyTorch; ASIC & RTL Design, Hardware–Software Co-design, Computer Architecture, VLSI Design, Logic Synthesis, Parallel Programming

Experience

ASIC Design Intern — NVLink — May 2026 – August 2026

NVIDIA — Santa Clara, CA

- Designed **production Verilog modules** for the **NVLink Chip-to-Chip interconnect**, implementing **packet framing/deframing**, **interconnect fanout reduction**, and **asynchronous/synchronous transaction** handling
- Implemented **CDC (Clock Domain Crossing)** logic for high-speed chip-to-chip links, applying synchronizer chains and handshake protocols to ensure **functional correctness** and **timing closure** across async clock boundaries
- Participated in an internal NVIDIA Hackathon to optimize a **Nemotron** model for efficient execution on an **LPU**, earning **13th place** among competing teams

Research Assistant — August 2026 – Present

PlatformX — Champaign, IL

- Helped develop an **architectural simulator** for **AI wafer-scale accelerators**, enabling **hardware–software co-design** studies that identify improvements in **bandwidth utilization** and **compute efficiency** through reconfigurable designs
- Modeled **AI/ML workload dataflow** across accelerator subsystems to evaluate **performance–area–power tradeoffs**, providing architectural feedback to guide **compute and memory subsystem** design decisions
- Extended **GPGPU-Sim** (cycle-accurate simulator) with **C++** to enable detailed profiling of **memory access patterns** and **interconnect bandwidth**; applied across **GPU Processing Cluster (GPC)** designs studying **L1 cache slice** and **SM** interactions optimized for **AI workloads**

Computer Architecture (ECE 411) Teaching Assistant — August 2025 – Present

UIUC Computer Organization and Design — Champaign, IL

- Trained students in designing and implementing **out-of-order RISC-V processors** with **non-blocking pipeline caching**, **branch prediction & recovery**, out-of-order **load/store units**, and **superscalar** execution
- Mentored students on **RTL debugging** across pipeline and cache projects using **Verdi** and **Verilator**; orchestrated competition profiling **30+ RISC-V processors** on **Power, Performance, and Area (PPA)** metrics

Projects

FORTE: Secure Processor Tape-Out — April 2025 – Present

Silicon Research Project (65nm Tape-Out) — SystemVerilog, VCS Design Compiler, Verdi, Spyglass, Yosys

- Designed and implemented **fault-tolerance RTL** on a **CORE-V Wally** (RV64IMA, Sv39 MMU, PMP, M/S-mode) baseline targeting **65nm tape-out**; features include **SECDED ECC** across register file, L1 I/D-cache, PTW/TLB, a **cacheline and CSR scrubbing FSM**, and protection against EMI Instruction/Data Exfiltration
- Implemented **TMR hardening** for security-critical CSRs (`mstatus`, `mepc`, `mtvec`, PMP regs) to defend against physical fault-injection attacks
- Conducted area feasibility study using **VCS Design Compiler** using TSMC's 65nm PDK; got pre-PnR area to 0.63 mm^2 at an operating frequency of 250MHz

RV-32IM Out-of-Order RISC-V Processor — January 2025 – May 2025

Computer Architecture Project — SystemVerilog, C++

- Designed and verified a pipelined, out-of-order, **N-way superscalar processor** based on the **RV32IM ISA**, with explicit register renaming
- Implemented a split Load/Store Unit for **speculative memory operations**, branch predictors, and **non-blocking pipelined cache**, including **functional coverage** metrics to assess verification completeness
- Debugged RTL with waveform tracing using **Verilator** and **Verdi**; verification involved writing unit-level **SystemVerilog testbenches** and assertions to validate pipeline stages, and simulations were performed with industry tools (VCS-compatible)
- Performed synthesis and power-performance-timing (**PPA**) analysis using **Synopsys Design Compiler**; evaluated across multiple testcases for critical path delays, area constraints, and **dynamic power** efficiency